Suppose $(X, \mathscr{U})$ is a uniform space. We say that X is totally bounded if for all $D \in \mathscr{U}$ there exists a finite set $S \subseteq X$ such that $X = D[S]$.

Remark: If $\mathscr{B}$ is a filterbase for $\mathscr{U}$ then X is totally bounded iff for all $D \in \mathscr{B}$ we can find a finite set $S \subseteq X$ such that $X = D[S]$. From this observation, hopefully it is clear how this definition of total boundedness agrees with the metric space definition when we equip our metric space with the metric uniform structure.

Proposition: Suppose $(X, \mathscr{U}_X)$ and $(Y, \mathscr{U}_Y)$ are uniform spaces and $\phi : X \to Y$ is a uniformly continuous surjection. If X is totally bounded then so is Y.

Proof:
Consider any $E \in \mathscr{U}_Y$ and then let $D := (\phi \times \phi)^{-1}(E) \in \mathscr{U}_X$. Since X is totally bounded, we can find a finite set $S \subseteq X$ with $D[S] = X$. In turn, we have that $Y = E[\phi(S)]$. After all, suppose $y \in Y$. Since ϕ is surjective, there exists $x \in X$ such that $\phi(x) = y$. Yet now pick $s \in S$ such that sDx. Then $\phi(s)Ey$. ■

Corollary: If X and Y are uniform spaces, and $\phi : X \to Y$ is a uniform equivalence, then X is totally bounded iff Y is totally bounded.

One of the main reasons for caring about uniform spaces is that they let us define what it means for a sequence to be Cauchy.

Suppose $(X, \mathscr{U})$ is a uniform space and $\{x_n\}_{n\in\mathbb{N}}$ is a sequence in X. Then we say $\{x_n\}_{n\in\mathbb{N}}$ is a Cauchy sequence if:

$$\forall D \in \mathscr{U},\ \exists N \in \mathbb{N}\ s.t.\ \forall n, m \geq N,\ x_n D x_m$$

Remark: Like with total boundedness, if $\mathscr{B}$ is a filterbase for $\mathscr{U}$ then it suffices to check the condition for a sequence being Cauchy only for the elements in $\mathscr{B}$. With this in mind, hopefully it is clear how this definition of a sequence being Cauchy agrees with the metric space definition when we equip out metric space with the metric uniform structure.

Similarly, we say a net $\langle x_\alpha \rangle_{\alpha\in A}$ is Cauchy if:

$$\forall D \in \mathscr{U},\ \exists \alpha \in A\ s.t.\ \forall \beta, \gamma \geq \alpha,\ \beta D \gamma$$

Once we define a topology associated with a uniform structure, we'll be able to discuss whether a uniform space is complete or not (i.e. if all Cauchy sequences and nets converge or not). Furthermore, we'll be able to discuss taking completions of uniform spaces.

Proposition: Suppose $(X, \mathscr{U}_X)$ and $(Y, \mathscr{U}_Y)$ are uniform spaces, $\langle x_\alpha \rangle_{\alpha\in A}$ is a Cauchy net in X, and $\phi : X \to Y$ is a uniformly continuous map. Then $\langle \phi(x_\alpha) \rangle_{\alpha\in A}$ is a Cauchy net in Y.

Proof:
Let E be a relation in $\mathscr{U}_Y$.

By the way, something I've not mentioned before is that a relation in $\mathscr{U}_Y$ would normally be called an entourage. The reason I wasn't using that terminology before is that the word "entourage" makes my mind immediately think of a neighborhood in Y as opposed to a subset of $Y \times Y$. That said, it would be helpful to not have to keep explicitely naming the uniform structure of X and Y. So, I'm going to start calling relations in uniform structures "entourages".

Since ϕ is uniformly continuous, we know that $D := (\phi \times \phi)^{-1}(E)$ is an entourage on X. In turn, as $\langle x_\alpha \rangle_{\alpha \in A}$ is a Cauchy net, we know there exists $\alpha_0 \in A$ with $x_\beta D x_\gamma$ for all $\beta, \gamma \geq \alpha$. And now $\phi(x_\beta) E \phi(x_\gamma)$ for all $\beta, \gamma \geq \alpha$. Since E was arbitrary, we can conclude that $\langle \phi(x_\alpha) \rangle_{\alpha \in A}$ is a Cauchy net in Y. ■

Proposition: Suppose X is a uniform space. Then X is totally bounded if and only if every net $\langle x_\alpha \rangle_{\alpha \in A}$ in X admits a Cauchy subnet.